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Global Warming and Winter Snowstorms: The Moisture Paradox

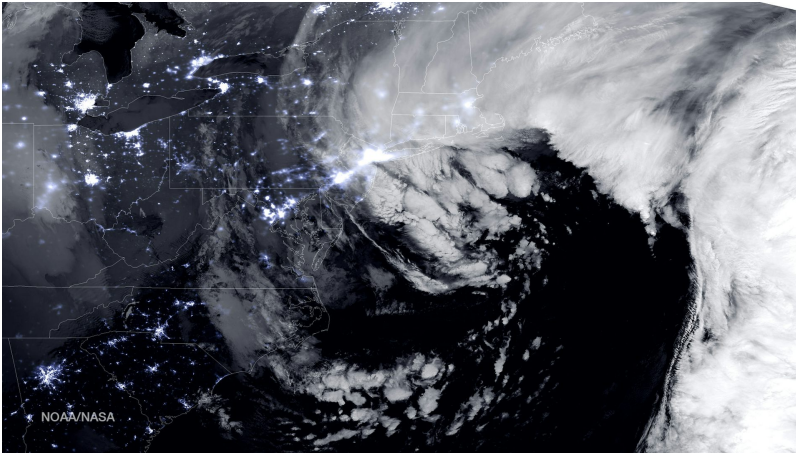
Abstract

If the planet is warming, why are some snowstorms getting bigger? In early 2015, after a blizzard buried parts of New England under more than three feet of snow, climate scientist Kevin Trenberth explained the physics. Warmer oceans and air add moisture to winter storms, so mid-winter snowfalls can grow heavier, even as the snow season shrinks at both ends.

Introduction

At first glance, asking whether global warming causes more snow seems silly: if it gets warm enough, there is no snow. Some people have even used big snowstorms to cast doubt on climate change. But to understand the connection, scientists look at the conditions that produce the heaviest snowfalls and how climate change affects them. There is a saying that it can be too cold to snow. It is a myth, but it has a basis in fact: very cold air is dry, and snow falls as tiny, light crystals. The heaviest snowfalls happen when surface temperatures are just below freezing, from about 28°F to 32°F. These Goldilocks conditions are becoming more likely in mid-winter.

How Do Scientists Study This?

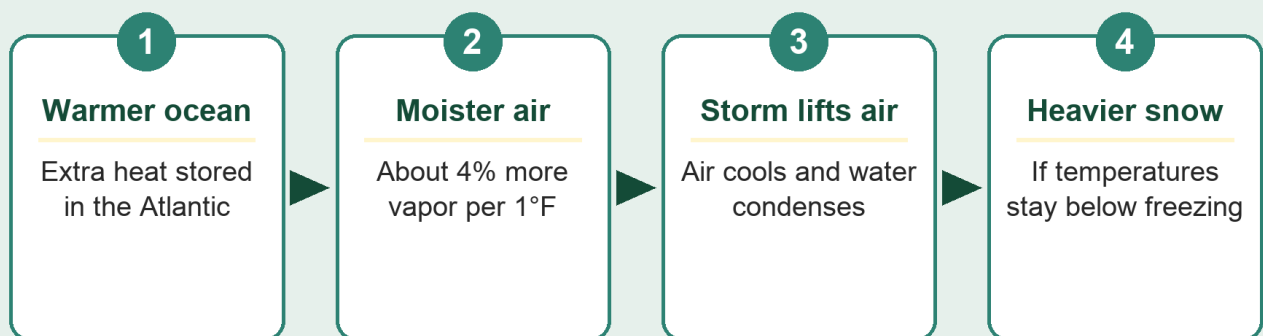


FROM THE ARTICLE

The first blizzard of 2015 viewed from space. Image: NOAA/NASA, CC BY.

Trenberth's explanation rests on a basic physical law: the maximum amount of moisture air can hold increases exponentially with temperature. Near sea level, air can hold about 4% more moisture for each 1°F of warming. Precipitation forms when storms lift moist air, which expands, cools, and condenses into clouds, rain, or snow. Since the main source of a storm's precipitation is moisture already in the atmosphere, more moisture means heavier snow or rain. He combined this physics with observations: sea surface temperature records, Northern Hemisphere snow cover data, and case studies of the 2010 Snowmageddon storm and the January 2015 blizzard.

How warming can mean more snow



Study the Process

Apply the physics

Air holds about 4% more moisture for every 1°F of warming, fueling heavier precipitation.

Check the oceans

Sea surface temperature records show how much extra warmth and moisture storms can tap into.

Compare seasons

Snow cover observations reveal changes in mid-winter versus spring snow.

What Did They Discover?

Carbon dioxide has risen over 40% since about 1900, and over 90% of the resulting extra heat has gone into the oceans. Global sea surface temperatures have risen about 1°F since 1970, and air over the oceans is more than 1°F warmer and 5% moister. In the North Atlantic, waters were over 2°F above average across more than 1,000 miles. Storms forming along the sharp contrast between cold continental air and this warm, moist ocean air can dump more snow. Snowfall at 32°F can be at least double that at 14°F. Snow cover shows slight increases in mid-winter but huge losses in spring, so the snow season is shrinking even as individual storms grow.

Snow and warming by the numbers

4%

more moisture air can hold per 1°F of warming

2×

snowfall at 32°F compared with 14°F, at least

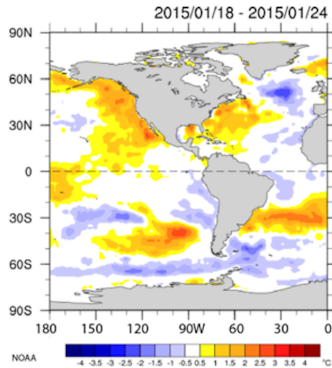
3 ft+

of snow in some areas in the January 2015 blizzard

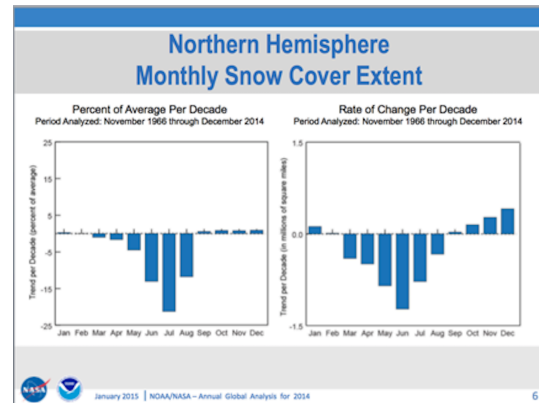
During the 2010 Snowmageddon storm, subtropical Atlantic waters were up to 3°F (1.5°C) above normal, feeding extraordinary moisture into the storm.

Look at the Evidence

FROM THE ARTICLE



Sea surface temperature anomalies, January 18-24, 2015, relative to 1981-2010. Image: NOAA.



Northern Hemisphere snow cover trends by month, 1966-2014: slight mid-winter gains, large spring losses. Image: NOAA/NASA.

Takeaways

Warmer air, more moisture

Each 1°F of warming lets air hold about 4% more water vapor.

Bigger mid-winter storms

Near-freezing storms fed by warm oceans can drop heavier snow.

Shorter snow season

Early and late winter turn to rain, so spring snow cover is shrinking.

Key Terms

Water-holding capacity

The maximum amount of water vapor air can contain, which increases rapidly as temperature rises.

Condensation

When water vapor cools and turns into liquid droplets or ice crystals, forming clouds and precipitation.

Extra-tropical storm

A large storm that forms outside the tropics along the boundary between warm and cold air masses.

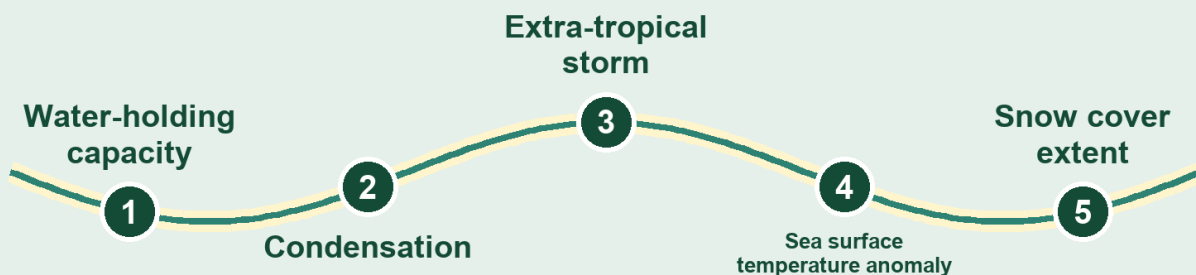
Sea surface temperature anomaly

The difference between the current ocean surface temperature and its long-term average.

Snow cover extent

The total area of land covered by snow, measured by satellites.

Vocabulary Path · Global Warming and Winter Snowstorms



Activity

The Goldilocks Snow Graph

Students graph how much moisture air can hold at different temperatures and find the snowstorm sweet spot.

Materials: Graph paper; Calculator; Moisture values from the lesson

1. Use the lesson's data: air at 50°F holds about twice the moisture of air at 32°F, and air at 14°F holds only 24% of the 50°F value.
2. Set 50°F to 100 units. Calculate and plot the values for 32°F and 14°F.
3. Shade the range from 28°F to 32°F. Explain why this range produces the heaviest snow.
4. Use the 4% per °F rule to estimate the extra moisture if a storm's air warmed from 28°F to 30°F.
5. Write a short answer to the question: can global warming mean more snow?

References

Trenberth, K. (2015). Does global warming mean more or less snow? The Conversation.
<https://theconversation.com/does-global-warming-mean-more-or-less-snow-36936>